

Amit Kumar Mohanty

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Target Role: AI Developer I | HCLTech

PROFESSIONAL SUMMARY

AI/ML Engineer and Computer Science graduate (AI/ML specialization) with 1+ year of hands-on experience building Generative AI (GenAI) applications and deploying models in cloud-native environments. Skilled in Python, GenAI frameworks and LLM APIs (OpenAI, Gemini, LangChain), prompt engineering, and containerized microservices (Docker, Kubernetes, CI/CD). Proven ability to take solutions from ideation to MVP within Agile/Scrum teams. Solution-oriented, strong communicator, and eager to pursue AI/ML and cloud certifications from AWS, Google, Microsoft, and Red Hat.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 - July 2028

M.Tech in Artificial Intelligence and Machine Learning (Pursuing)

K.R. Mangalam University, Gurgaon

July 2021 - July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML)

EXPERIENCE

Research Associate - AI, Keywords Studios India - Gurgaon, Haryana

May 2025 - Present

- Trained and optimized enterprise-grade Generative AI and agentic AI models to autonomously execute complex multi-step workflows, improving task-completion reliability across cloud-based deployments.
- Contributed to a web-automation AI agent that interprets on-screen content and, from natural-language prompts, plans and executes browser actions to complete multi-step tasks.
- Built custom data pipelines and delivered 85% of self-generated training datasets to production, applying data preprocessing and feature engineering for high-quality model inputs.
- Awarded "Rookie of the Year 2025" for surpassing all KPIs through model training, data-pipeline optimization, and cross-functional collaboration in an Agile team.

Artificial Intelligence Intern, Infosys - Remote

Nov 2024 - Feb 2025

- Engineered an AI call-center system with LangChain and LangSmith, boosting model response accuracy by 35% and improving customer-AI interaction quality.
- Designed and A/B-tested prompt-engineering strategies with automated evaluation, reducing hallucinations by 20% and mitigating biased/undesired model outputs.
- Developed and deployed LLM features that reduced system latency by 25%, streamlining the model deployment process within a 12-engineer team.

PROJECTS

Dynamic Screen Companion

June 2025 - Present

- Built a real-time multi-modal GenAI assistant integrating the Gemini API to analyze live screen activity and deliver continuous voice feedback.
- Engineered a high-performance FastAPI microservice backend handling concurrent inference streams, containerized with Docker and CI/CD for cloud-native deployment.
- Optimized OCR and NLP pipelines (tokenization, embeddings) to maintain 95%+ extraction accuracy across varied screen layouts.

Diabetic Optiscan (Peer-reviewed research)

Jan 2025 - June 2025

- Built an AI tool detecting early diabetic retinopathy from retinal images; co-authored a peer-reviewed paper on the design and accuracy gains.
- Led integration of a Vision Transformer (ViT-B/16) on the APTOS dataset, optimizing self-attention to capture complex spatial patterns.
- Designed a custom wavelet technique that boosted detection precision by 50% over legacy methods.

SKILLS

Languages: Python, Java (basic), C++, SQL, R

Generative AI & NLP: LLMs, GenAI, OpenAI API, Gemini API, LangChain, LangGraph, RAG, Prompt Engineering, Fine-tuning (LoRA/QLoRA), Transformers, Tokenization, Embeddings, Hugging Face, BERT, AI Ethics & Bias Mitigation

Cloud-Native & DevOps: Docker, Kubernetes, Microservices, CI/CD, 12-Factor Apps, MLflow, Model Serving & Monitoring, Git, Agile/Scrum

Data & Cloud: AWS, GCP (Vertex AI), Data Pipelines, Feature Engineering, Vector Databases (FAISS/Chroma), Pandas, NumPy, MySQL, MongoDB, Power BI

Frameworks: FastAPI, TensorFlow, PyTorch, Scikit-learn, OpenCV

CERTIFICATIONS

- Ducat India - Data Science Professional Training (Python, ML, statistics, predictive modeling), Apr 2024 - Apr 2025.
- ISRO - AI/ML for Geodata Analysis (applied ML/DL to remote-sensing imagery).
- Actively pursuing cloud & AI/ML certifications (AWS, Google Cloud, Microsoft Azure, Red Hat).